



CHANDIGARH UNIVERSITY

CENTRE FOR DISTANCE & ONLINE EDUCATION

MAJOR PROJECT REPORT (UID : O24MCA110853)

Retail Marketing Optimization Systems

A project report submitted in fulfillment of the requirements for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

(Fourth Semester)

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CERTIFICATE

This is to certify that the Major Project entitled "**Retail Marketing Optimization Systems**" submitted by **Nehal Ojha** (Roll No: **O24MCA110853**) in fulfillment of the requirements for the award of the degree of **Master of Computer Applications** from **Chandigarh University**, Centre for Distance & Online Education, is a record of bona fide project work carried out under my supervision and guidance.

The matter embodied in this report has not been submitted anywhere else for the award of any other degree or diploma to the best of my knowledge and belief. This project fulfills all software engineering methodologies, systemic modeling, and implementation guidelines specified by the university.

Nehal Ojha

Candidate Signature

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DECLARATION

I, **Nehal Ojha**, student of **Master of Computer Applications (MCA)**, Fourth Semester, bearing Roll Number **O24MCA110853** at **Chandigarh University**, Centre for Distance & Online Education, hereby declare that the project work presented in this report entitled "**Retail Marketing Optimization Systems**" is an authentic work carried out by me individually under the guidance of my project supervisor.

I further declare that the source code, implementation parameters, dataset schemas, and software architecture diagrams included in this report have been custom-developed by me. All information, libraries, and frameworks utilized from third-party open-source ecosystems have been properly acknowledged and cited in the bibliography.

Date: June, 2026

Place: Hooghly , West Bengal

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ACKNOWLEDGEMENT

I express my sincere gratitude to my respected project guide for continuous encouragement, valuable suggestions, technical guidance, and constant support throughout the development of this project.

I also thank the Head of the Department and all faculty members of the Department of Computer Applications for providing the necessary facilities and academic environment required for completing this work.

I would like to acknowledge the open-source community, whose learning resources, datasets, and development tools made this project possible. Special thanks are extended to the creator of the reference analytics project that inspired the implementation and helped demonstrate a practical end-to-end data analytics workflow.

Finally, I express my heartfelt gratitude to my parents and friends for their encouragement and moral support during the completion of this project.

Nehal Ojha

ABSTRACT

Modern businesses generate large volumes of customer transaction data every day. However, transforming this raw information into meaningful business intelligence requires a systematic data analytics process. This project presents an end-to-end customer shopping behaviour analysis system developed using Python, SQL, and Power BI.

The proposed system performs data cleaning, pre-processing, exploratory data analysis, database integration, SQL-based business analysis, and interactive dashboard development. Python is used to prepare and clean the dataset before storing it in a relational database. SQL queries are employed to answer important business questions related to customer demographics, purchasing behavior, spending habits, and loyalty. Finally, Power BI is used to build interactive dashboards that help business managers identify trends and make informed decisions.

The developed system demonstrates how multiple data analytics technologies can be integrated into a unified workflow to extract actionable insights from retail customer data. The project also highlights best practices in data visualization, business intelligence, and decision support systems.

Keywords: Data Analytics, SQL, Python, Power BI, Customer Behaviour, Business Intelligence, Dashboard, Data Visualization.

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1. INTRODUCTION

1.1 Introduction

The rapid growth of digital commerce has resulted in organizations collecting enormous volumes of customer-related information. Every purchase, website visit, product search, and customer interaction contributes to a growing repository of data. While this information holds significant business value, its true potential can only be realized through systematic analysis and interpretation.

Data analytics has emerged as one of the most influential technologies for transforming raw data into meaningful business insights. Modern organizations rely heavily on analytical techniques to understand customer behavior, predict future trends, improve operational efficiency, and support strategic decision-making. Companies such as Amazon, Walmart, Flipkart, and many other retail organizations extensively use customer analytics to enhance user experience, optimize inventory management, and increase profitability.

This project focuses on developing an end-to-end customer shopping behavior analysis system using three widely adopted technologies: Python, SQL, and Power BI. Python is employed for data preprocessing, cleaning, and exploratory analysis. SQL facilitates efficient storage and retrieval of structured data while enabling the execution of complex business queries. Power BI serves as the visualization platform, presenting analytical findings through interactive dashboards that support data-driven decision-making.

The project demonstrates the complete workflow typically followed by professional data analysts, from importing raw data to presenting actionable business insights through dashboards.

1.2 Background of Study

Customer behavior analysis is the process of examining purchasing patterns, preferences, demographic characteristics, and transactional history to understand consumer decision-making. Businesses increasingly depend on such analyses to improve customer satisfaction, identify high-value customers, and optimize marketing strategies.

Traditional reporting methods often fail to provide timely insights due to their dependence on manual data processing. Contemporary data analytics platforms integrate programming languages, database systems, and visualization tools to automate analytical workflows. Python provides robust libraries for statistical analysis and data manipulation, SQL enables efficient querying of structured datasets, and Power BI offers advanced visualization capabilities that transform analytical results into interactive reports.

This integration significantly enhances an organization's ability to monitor key performance indicators (KPIs), identify emerging trends, and respond quickly to changing customer behavior.

1.3 Problem Statement

Retail organizations generate large quantities of customer transaction data every day. Despite having access to such information, many businesses struggle to transform raw datasets into actionable knowledge due to inadequate analytical infrastructure, inconsistent data quality, and limited visualization capabilities.

The absence of integrated analytics often results in:

- Difficulty identifying profitable customer segments.
- Limited understanding of purchasing behavior.
- Inefficient marketing campaigns.
- Poor inventory planning.
- Reduced customer retention.
- Delayed managerial decision-making.

This project addresses these challenges by implementing a comprehensive analytics workflow that combines Python, SQL, and Power BI to generate meaningful business intelligence.

1.4 Objectives

The primary objectives of this project are:

- To develop an end-to-end customer behavior analysis system.
- To preprocess and clean customer shopping data using Python.
- To perform exploratory data analysis for identifying patterns and trends.
- To store processed data in a relational SQL database.
- To answer business-oriented questions using SQL queries.
- To develop interactive dashboards using Power BI.
- To assist decision-makers in identifying profitable customer segments.
- To provide meaningful visualizations for strategic planning.

1.5 Scope of the Project

The scope of the project includes:

- Importing raw customer shopping data.
- Data preprocessing and cleaning.
- Handling missing and duplicate records.
- Exploratory Data Analysis.
- SQL database creation.
- Business query implementation.
- Interactive dashboard development.
- Business insight generation.
- Performance visualization.

The project does not include predictive machine learning models or real-time streaming analytics; its focus is descriptive and diagnostic analytics.

1.6 Significance of the Project

This project demonstrates the practical application of modern data analytics tools in solving real-world business problems. It enables organizations to gain a deeper understanding of customer behavior and purchasing trends, thereby supporting informed strategic decisions.

Academically, the project provides hands-on experience with Python, SQL, and Power BI while illustrating a complete analytics lifecycle suitable for industry-oriented learning.

1.7 Project Workflow

The project follows the workflow below:

1. Data Collection
2. Data Import using Python
3. Data Cleaning
4. Exploratory Data Analysis
5. SQL Database Creation
6. Data Loading into SQL
7. SQL-Based Business Analysis
8. Power BI Dashboard Development
9. Business Insight Generation
10. Report Preparation

1.8 Organization of report

The report is organized into the following chapters:

- **Chapter 1:** Introduction
- **Chapter 2:** Literature Review
- **Chapter 3:** Software Requirement Specification
- **Chapter 4:** System Design and Database Design
- **Chapter 5:** Project Implementation
- **Chapter 6:** Dashboard Development and Business Insights
- **Chapter 7:** Testing and Results
- **Chapter 8:** Conclusion and Future Scope

2. COMPREHENSIVE LITERATURE REVIEW

2.1 Introduction

The rapid advancement of digital technologies has transformed the way organizations collect, store, and analyze customer information. Every transaction, product search, online purchase, and customer interaction generates valuable data that can be analyzed to improve business performance. Data analytics has therefore become one of the most significant technological domains for modern organizations. It enables companies to identify customer preferences, forecast demand, optimize marketing strategies, and improve customer satisfaction.

Retail industries are increasingly adopting data-driven decision-making to remain competitive. Instead of relying solely on intuition, businesses use statistical analysis, database systems, and visualization tools to understand customer behavior. Technologies such as Python, SQL, and Power BI have become industry standards due to their efficiency, flexibility, and scalability.

This chapter reviews previous research in customer behavior analysis, business intelligence, data visualization, database management systems, and exploratory data analysis. The review establishes the academic foundation for the proposed system and demonstrates how modern analytical tools contribute to solving real-world business problems.

2.2 Customer Behavior Analytics

Customer behavior analytics refers to the systematic study of customer interactions, purchasing habits, preferences, and decision-making processes. Organizations analyze these patterns to understand what motivates customers to purchase products and how they interact with various services.

Behavioral analysis generally includes:

- Purchase frequency analysis
- Customer segmentation
- Spending pattern analysis
- Product preference identification
- Seasonal buying trends
- Customer loyalty evaluation
- Revenue contribution analysis

Retail organizations utilize these analytical methods to personalize recommendations, optimize inventory management, improve marketing campaigns, and increase customer retention.

2.3 Data Analytics in retail industry

The retail industry generates enormous quantities of structured and unstructured data every day. Effective utilization of this information enables businesses to improve operational efficiency while maximizing profitability.

Major applications include:

- Sales forecasting
- Customer segmentation
- Inventory optimization
- Product recommendation systems
- Customer lifetime value analysis
- Market basket analysis
- Fraud detection
- Promotional campaign analysis

Modern retail companies increasingly depend upon business intelligence platforms for real-time monitoring of organizational performance.

2.4 Python for Data Analytics

Python has become the most widely used programming language in the field of data analytics due to its simplicity and extensive ecosystem of libraries.

Important Python libraries include:

Pandas

Pandas provides powerful data structures for handling tabular data. It supports filtering, grouping, aggregation, missing value handling, duplicate removal, merging, and statistical operations.

Applications:

- Data cleaning
- Data preprocessing
- Dataset transformation
- Feature engineering

NumPy

NumPy offers high-performance numerical computation through multidimensional arrays.

Applications:

- Mathematical operations
- Matrix computation
- Statistical calculations

Matplotlib

Matplotlib is widely used for creating visual representations of data.

Common charts include:

- Line graphs
- Histograms
- Scatter plots
- Pie charts
- Bar graphs

Seaborn

Seaborn extends Matplotlib by providing aesthetically pleasing statistical graphics.

Used for:

- Correlation heatmaps
- Pair plots
- Distribution plots
- Box plots
- Violin plots

2.5 SQL for Business Analytics

Structured Query Language (SQL) is the standard language for managing relational databases. SQL enables analysts to retrieve meaningful information from structured datasets using efficient query operations.

Major SQL operations include:

- SELECT
- INSERT
- UPDATE
- DELETE
- GROUP BY
- ORDER BY
- HAVING

- Aggregate Functions
- Window Functions
- Joins
- Subqueries

Business organizations utilize SQL to generate reports regarding sales performance, customer segmentation, employee productivity, financial transactions, and inventory management.

2.6 Business Intelligence using PowerBI

Power BI is a business intelligence platform developed by Microsoft that converts raw datasets into interactive dashboards and reports.

Major features include:

- Interactive Dashboards
- KPI Cards
- Slicers
- Drill-through Analysis
- Geographic Maps
- Time-Series Charts
- Forecasting
- Report Sharing

Power BI enables business managers to monitor organizational performance in real time through visually intuitive reports.

2.7 Exploratory Data Analysis (EDA)

EDA is the process of understanding a dataset before performing advanced analysis.

Common EDA activities include:

- Missing value identification
- Duplicate detection
- Statistical summary
- Outlier analysis
- Correlation analysis
- Distribution analysis
- Trend identification

EDA ensures data quality and improves the reliability of subsequent analytical models.

2.8 Existing Systems

Traditional retail reporting systems primarily generate static reports that require manual interpretation.

Limitations

- Static reports
- Lack of visualization
- Delayed reporting
- Limited interactivity
- Poor scalability
- Manual analysis

2.9 Proposed System

The proposed system integrates Python, SQL, and Power BI into a unified analytical workflow.

Advantages include:

- Automated preprocessing
- Efficient SQL querying
- Interactive visualization
- Business insight generation
- Better decision support
- Improved reporting efficiency

2.10 Research Gap

Despite significant progress in business intelligence technologies, many small and medium-sized enterprises continue to face challenges in implementing complete analytical workflows.

Common research gaps include:

- Lack of integrated analytics
- Poor visualization
- Inadequate data preprocessing
- Limited automation
- Absence of standardized workflows

The proposed project addresses these issues by developing a complete end-to-end analytics pipeline.

3. SOFTWARE REQUIREMENT SPECIFICATION (SRS)

3.1 Introduction

A Software Requirement Specification (SRS) defines the functional and non-functional requirements of a software system. It serves as a blueprint for designing, developing, testing, and maintaining the application.

The proposed Customer Shopping Behavior Analysis System aims to provide an integrated platform for customer data analysis using Python, SQL, and Power BI.

3.2 Purpose

The purpose of the system is to:

- Import customer shopping data.
- Clean and preprocess datasets.
- Store processed information in SQL databases.
- Execute business queries.
- Generate interactive dashboards.
- Support business decision-making

3.3 Scope

The software covers the complete lifecycle of customer data analysis.

Major modules include:

- Data Import
- Data Cleaning
- Data Transformation
- SQL Database
- Business Analytics
- Dashboard Development
- Report Generation

3.4 Functional Requirements

REQ ID	DETAILED OPERATIONAL SCOPE & ALGORITHMIC BOUNDS
FR-1	Import customer dataset
FR-2	Validate dataset integrity
FR-3	Handle missing values
FR-4	Remove duplicate records
Fr-5	Perform Exploratory Data Analysis
FR-6	Create SQL Database

FR-7	Insert clean records
FR-8	Execute Analytical SQL queries
FR-9	Generate Power BI Dashboard
FR-10	Display Business Insights

3.5 Non-functional Requirements

REQ ID	DETAILED OPERATIONAL SCOPE & ALGORITHMIC BOUNDS
Performance	System should process large datasets efficiently
Reliability	System should produce consistent analytical result
Scalability	Future datasets can be incorporated without significant redesign.
Security	Database access should be restricted to authorized users.
Maintainability	Modules should be independently maintainable.
Probability	System should execute on Windows, Linux, and macOS environments where the required software is installed.

3.6 Hardware Requirements

Hardware Component	Recommended Specification
Processor Type	Intel Core i5 core and above
System Memory (RAM)	Minimum 8GB
Storage Capacity	255 GB SSD
Display	1366*768 or higher

3.7 Software Requirements

Software Component	Version
Python	3.x
Jupyter Notebook	latest
MySQL	8.x
Power BI Desktop	Latest
Windows	Windows 10/11

3.8 Feasibility study

Technical Feasibility

The required technologies are open-source or freely available for educational use and are widely adopted in the data analytics industry.

Economic Feasibility

Development cost is minimal because the project primarily uses free tools and an openly available dataset.

Operational Feasibility

The proposed system is easy to use and provides actionable insights through intuitive dashboards.

Schedule Feasibility

The project can be completed within a typical academic semester with proper planning.

3.9 Assumptions

- Input data is available in CSV format.
- Users have basic knowledge of SQL and Python.
- Power BI Desktop is installed.
- Database connectivity is properly configured.

3.10 Constraints

- The project focuses on descriptive analytics and dashboard reporting.
- Predictive machine learning models are outside the current scope.
- Dashboard functionality depends on the quality and completeness of the input data.

3.11 Risk Analysis

Potential risks include:

- Poor data quality.
- Database connection failures.
- Incomplete datasets.
- Incorrect SQL queries.
- Software compatibility issues.

Mitigation strategies involve data validation, regular backups, query testing, and environment verification.

4. SYSTEM ANALYSIS AND DESIGN

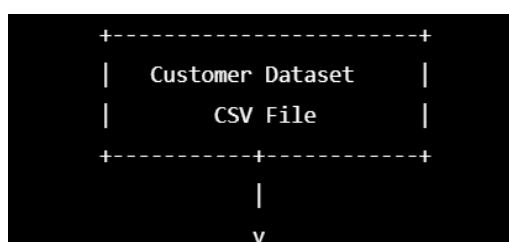
4.1 Introduction

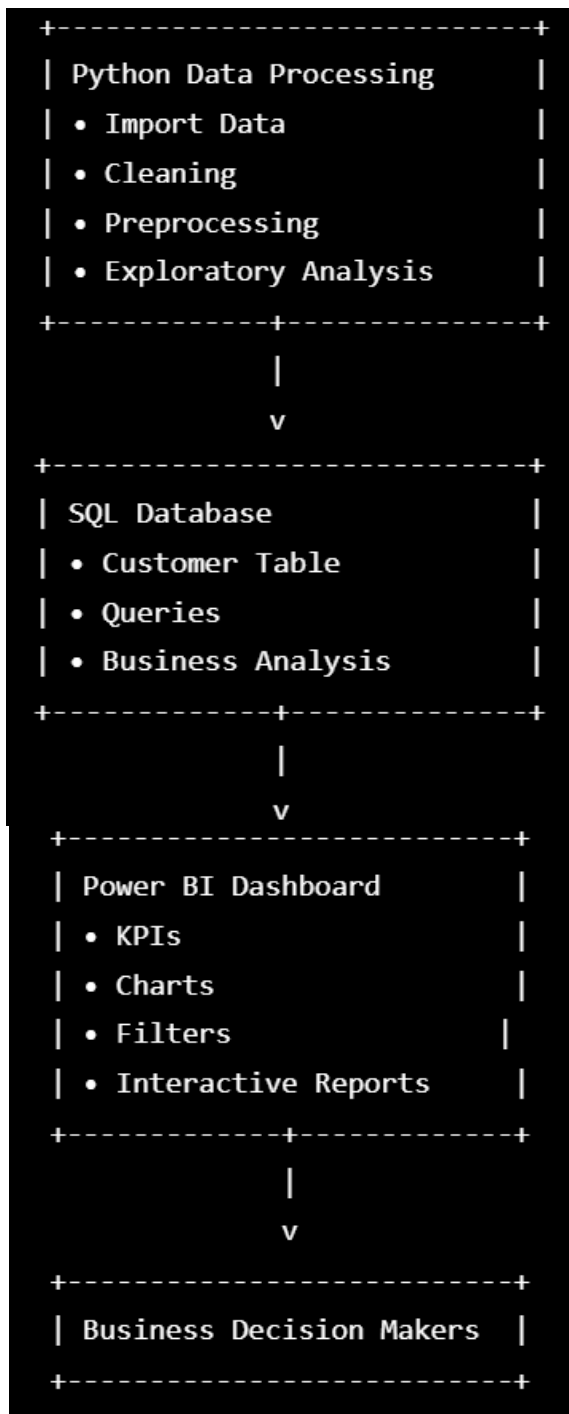
System analysis and design is a critical phase in software development that defines the overall structure, workflow, and interactions of a system before implementation. For this project, the design focuses on integrating data collection, preprocessing, SQL-based storage and analysis, and interactive visualization into one coherent analytics pipeline.

The objective is to ensure that each stage of the workflow transforms raw customer shopping data into meaningful business insights in a structured, maintainable, and scalable manner.

4.2 System Architecture

The proposed system follows a layered architecture consisting of five major components:



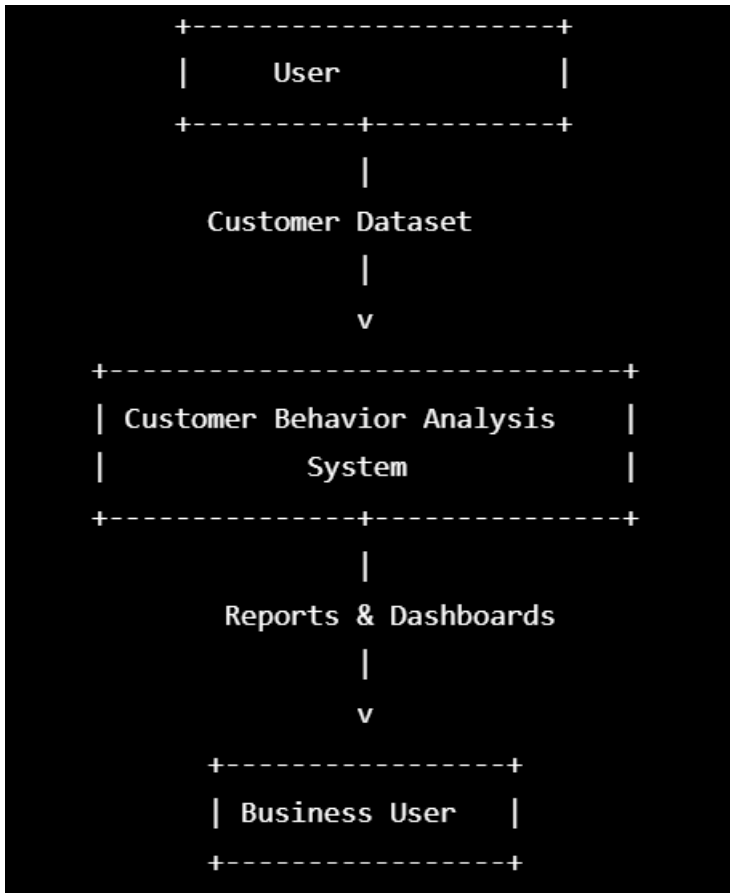


Explanation

1. **CSV Dataset** serves as the raw data source.
2. **Python** performs preprocessing, cleansing, and exploratory analysis.
3. The cleaned data is stored in a **SQL database**.
4. **SQL** is used to answer business-related questions through structured queries.
5. **Power BI** visualizes the analytical results through interactive dashboards.

4.3 Data Flow Diagram (DFD)

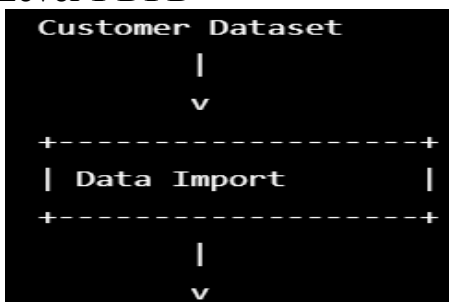
Level 0 DFD (Context Diagram)

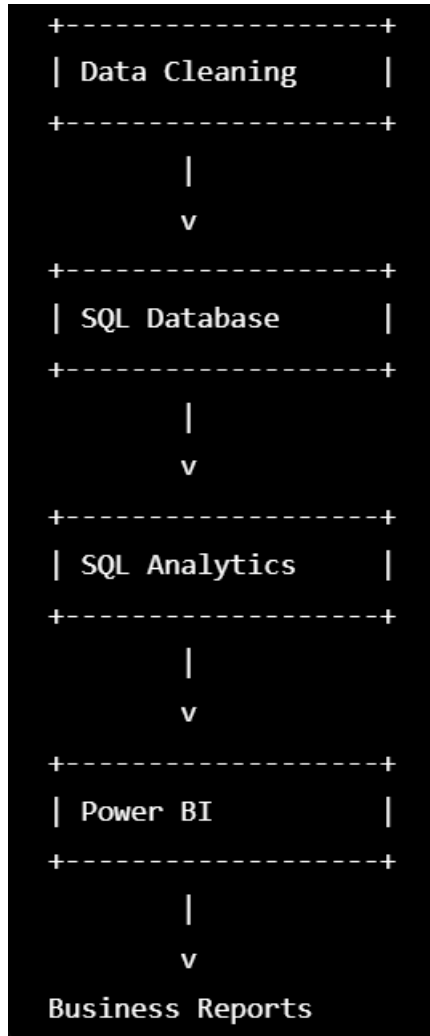


Description

The customer dataset is provided to the analytics system, which processes the data and produces dashboards and analytical reports for business users.

Level 1 DFD





4.4 Entity relationship (ER) Diagram

The customer behavior dataset primarily contains customer demographic details and shopping information. A simplified conceptual ER model is shown below.

```

+-----+
| CUSTOMER                                |
+-----+
| Customer_ID (PK)                       |
| Age                                    |
| Gender                                 |
| Location                               |
| Income                                |
+-----+
|                                         |
|                                         |
|                                         |
+-----V-----+
| PURCHASE                               |
+-----+
| Purchase_ID (PK)                       |
| Customer_ID (FK)                       |
| Category                               |
| Purchase_Amount                         |
| Payment_Method                         |
| Shopping_Mall                           |
| Date                                   |
+-----+

```

4.5 Database schema

CUSTOMER Table :

Field	Type
Customer_ID	INT
Age	INT
Gender	VARCH AR
Location	VARCH AR
Income	FLOAT

PURCHASE Table :

Field	Type
Purchase_ID	INT
Customer_ID	INT
Category	VARCHAR
Purchase_Amount	DECIMAL
Payment_Method	VARCHAR
Shopping_Mall	VARCHAR
Purchase_Date	DATE

4.6 Module Design

The proposed system consists of six independent modules.

Module 1

Data Import Module

Responsibilities:

- Import CSV
- Read dataset
- Validate records

Module 2

Data Cleaning Module

Responsibilities:

- Missing values
- Duplicate removal
- Incorrect values
- Data conversion

Module 3

Exploratory Data Analysis

Responsibilities:

- Statistical summary
- Distribution analysis
- Correlation
- Visualization

Module 4

SQL Analytics

Responsibilities:

- Store data
- Execute SQL
- Generate KPIs
- Customer segmentation

Module 5

Dashboard Module

Responsibilities:

- Interactive charts
- Filters
- KPI cards
- Drill-down reports

Module 6

Report Generation

Responsibilities:

- Business insights
- Dashboard interpretation
- Final reporting

5. SYSTEM IMPLEMENTATION

5.1 Importing Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Explanation:

- **Pandas** is used for loading and manipulating tabular data.
- **NumPy** provides numerical computation functions.
- **Matplotlib** generates charts and graphs.
- **Seaborn** creates advanced statistical visualizations.

These libraries provide the necessary tools for preprocessing, analysis, and visualization.

5.2 Loading the dataset

```
df = pd.read_csv("customer_shopping_behavior.csv")
```

Explanation:

The `read_csv()` function imports the CSV dataset into memory as a DataFrame. This DataFrame acts as the primary data structure used throughout the implementation.

Once loaded successfully, the dataset is ready for exploration and preprocessing.

5.3 Previewing the dataset

```
df.head()
```

Explanation:

The `head()` function displays the first five rows of the dataset. This helps verify that the dataset has been imported correctly and provides an overview of the available attributes.

Expected Output:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

5.4 Understanding Data structure

```
df.info()
```

Explanation: This function displays:

- Total number of records
- Total number of columns
- Data type of each column
- Number of non-null values
- Memory usage

The information obtained helps identify missing values and inappropriate data types that require preprocessing.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   object
3   Item Purchased                       3900 non-null   object
4   Category                            3900 non-null   object
5   Purchase Amount (USD)                3900 non-null   int64
6   Location                             3900 non-null   object
7   Size                                 3900 non-null   object
8   Color                                3900 non-null   object
9   Season                               3900 non-null   object
10  Review Rating                        3863 non-null   float64
11  Subscription Status                  3900 non-null   object
12  Shipping Type                        3900 non-null   object
13  Discount Applied                     3900 non-null   object
14  Promo Code Used                      3900 non-null   object
15  Previous Purchases                   3900 non-null   int64
16  Payment Method                       3900 non-null   object
17  Frequency of Purchases                3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

5.5 Statistical Summary

```
df.describe(include='all')
```

Explanation:

The statistical summary provides numerical measures such as:

- Mean
- Standard deviation
- Minimum value
- Maximum value
- Quartiles

For categorical variables, it displays the most frequent values and unique counts. These statistics provide an initial understanding of customer demographics and purchasing behaviour.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	NaN	6	7
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	NaN	677	584
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN

5.6 Missing Value Deletion

```
df.isnull().sum()
```

Explanation:

The `isnull()` function identifies missing entries in each column, while `sum()` counts the total missing values. During analysis, the **Review Rating** column was found to contain missing values that required preprocessing.

Identifying missing data at this stage ensures the reliability of subsequent analysis.

```
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    37
Subscription Status  0
Shipping Type    0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method   0
Frequency of Purchases  0
dtype: int64
```

5.7 Data Preprocessing

Data preprocessing is an essential step in data analytics because raw datasets often contain inconsistencies such as missing values, redundant columns, and non-standard naming conventions. In this project, preprocessing was carried out using the Pandas library to improve data quality before storing the dataset in the SQL database.

The preprocessing tasks performed include:

- Handling missing values
- Renaming column names
- Feature engineering
- Removing redundant attributes

5.8 Handling Missing Values

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(
    lambda x: x.fillna(x.median())
)
```

Explanation:

Instead of removing rows with missing values, the missing ratings were replaced using the **median review rating of the corresponding product category**.

This method was preferred because:

- It preserves all customer records.
- It avoids bias introduced by deleting observations.
- Median values are less affected by extreme ratings than the mean.

After replacing the missing values, the dataset was checked again to ensure that no null values remained.

```
df.isnull().sum()

Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    0
Subscription Status  0
Shipping Type    0
Discount Applied 0
Promo Code Used  0
Previous Purchases 0
Payment Method   0
Frequency of Purchases 0
dtype: int64
```

5.9 Renaming Column names

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(" ", "_")

df.rename(
    columns={
        "purchase_amount_(usd)": "purchase_amount"
    },
    inplace=True
)
```

Explanation

Renaming columns simplifies coding and prevents syntax errors caused by spaces or special characters. It also ensures compatibility with SQL table names and Power BI fields.

```
df.columns

Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

5.10 Feature Engineering

(a) Age Group Creation

The customer age was divided into four categories using quantile-based binning.

```
labels = [
    'Young Adult',
    'Adult',
    'Middle-aged',
    'Senior'
]

df['age_group'] = pd.qcut(
    df['age'],
    q=4,
    labels=labels
)
```

Explanation

The new **age_group** attribute simplifies demographic analysis and enables comparison of purchasing behavior across different age segments.

```
df[['age', 'age_group']].head(10)
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

(b) Purchase Frequency Conversion

The purchase frequency values were converted from categorical text into numerical values representing the approximate number of days between purchases.

```
frequency_mapping = {  
    'Weekly': 7,  
    'Bi-Weekly': 14,  
    'Monthly': 30,  
    'Quarterly': 90,  
    'Annually': 365  
}  
  
df['purchase_frequency_days'] = (  
    df['frequency_of_purchases']  
    .map(frequency_mapping)  
)
```

Explanation

The numerical representation enables statistical analysis, correlation studies, and visualization of purchase intervals.


```
df[['purchase_frequency_days', 'frequency_of_purchases']].head(10)
```

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

5.11 Removing Redundant Attributes

```
(df['discount_applied'] ==  
df['promo_code_used']).all()
```

```
df.drop(  
    'promo_code_used',  
    axis=1,  
    inplace=True  
)
```

Explanation

Removing duplicate information reduces storage requirements and simplifies the dataset without affecting analytical results.

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',  
      'purchase_amount', 'location', 'size', 'color', 'season',  
      'review_rating', 'subscription_status', 'shipping_type',  
      'discount_applied', 'previous_purchases', 'payment_method',  
      'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],  
      dtype='object')
```

5.12 Database Integration

After preprocessing, the cleaned dataset is stored in a relational database for structured querying and business analysis.

The project demonstrates database connectivity using SQLAlchemy with PostgreSQL (and similar logic can be applied for MySQL).

```
from sqlalchemy import create_engine

engine = create_engine(
    "postgresql://username:password@localhost:5432/customer_behavior"
)
```

Explanation

SQLAlchemy provides a convenient interface for establishing a database connection. The connection string specifies the database type, username, password, host, port, and database name.

Once connected, the processed DataFrame can be exported directly to SQL tables.

5.13 Exporting Data to SQL

```
df.to_sql(
    "customer_shopping",
    engine,
    if_exists="replace",
    index=False
)
```

Explanation

The `to_sql()` function automatically creates the table (if necessary) and inserts all processed records into the database. This eliminates manual data entry and ensures consistency between Python and SQL.

5.14 SQL Business Analysis

Query 1: Total Customers

```
SELECT COUNT(*) AS total_customers
FROM customer_shopping;
```

Purpose: Determines the total number of customer records.

Query 2: Average Purchase Amount

```
SELECT ROUND(  
    AVG(purchase_amount),2)  
FROM customer_shopping;
```

Purpose: Calculates the average spending per customer.

Query 3: Category wise sales

```
SELECT  
category,  
SUM(purchase_amount) AS total_sales  
FROM customer_shopping  
GROUP BY category  
ORDER BY total_sales DESC;
```

Purpose: Identifies the highest revenue-generating product categories.

Query 4: Payment method analysis

```
SELECT  
payment_method,  
COUNT(*)  
FROM customer_shopping  
GROUP BY payment_method;
```

Purpose: Shows the most frequently used payment methods.

Query 5: Customer Age group distribution

```
SELECT  
age_group,  
COUNT(*)  
FROM customer_shopping  
GROUP BY age_group;
```

Purpose: Analyzes the demographic distribution of customers.

5.15 Power BI Dashboard

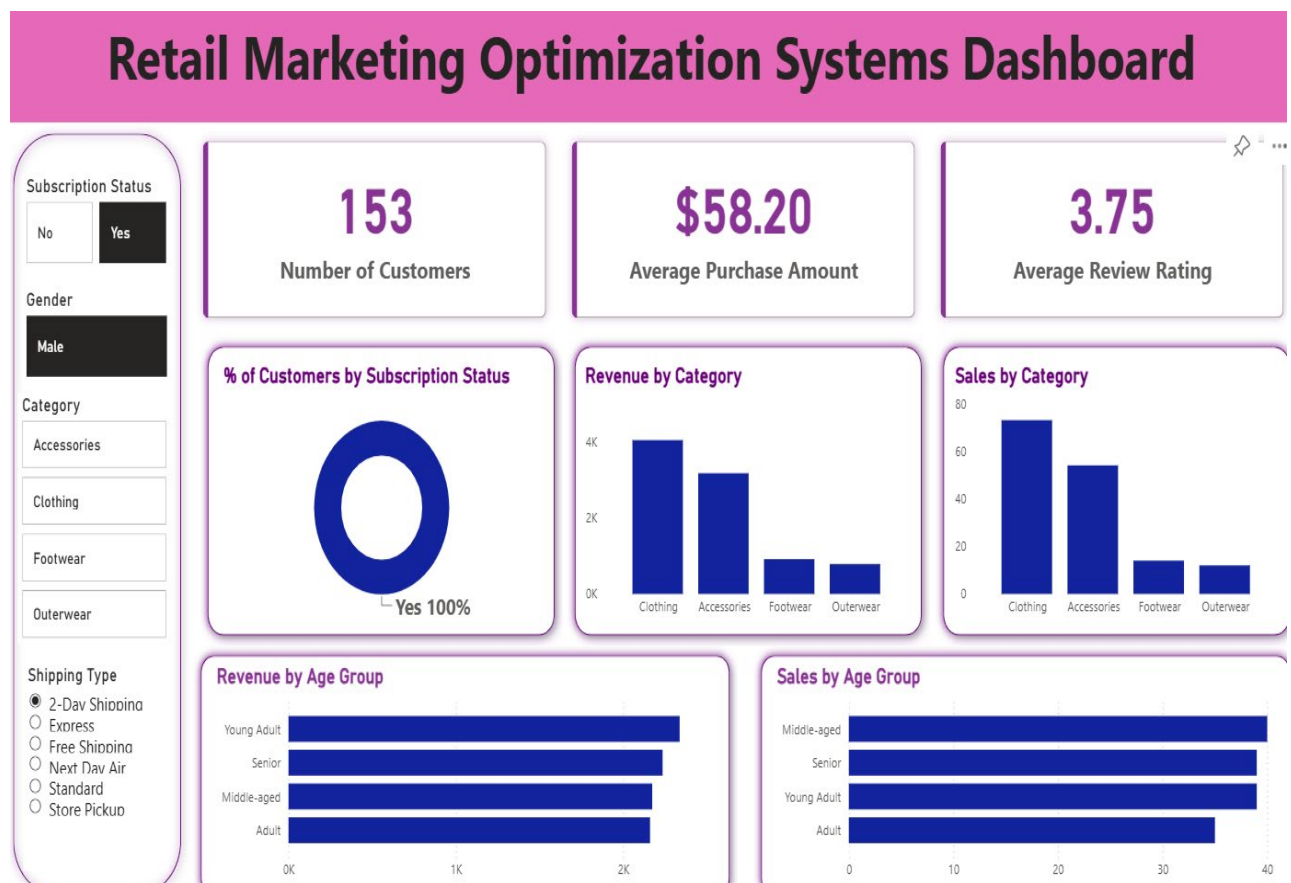
The processed SQL data is imported into Power BI to create an interactive dashboard. The dashboard enables users to explore customer behavior through visualizations and filters.

The dashboard includes:

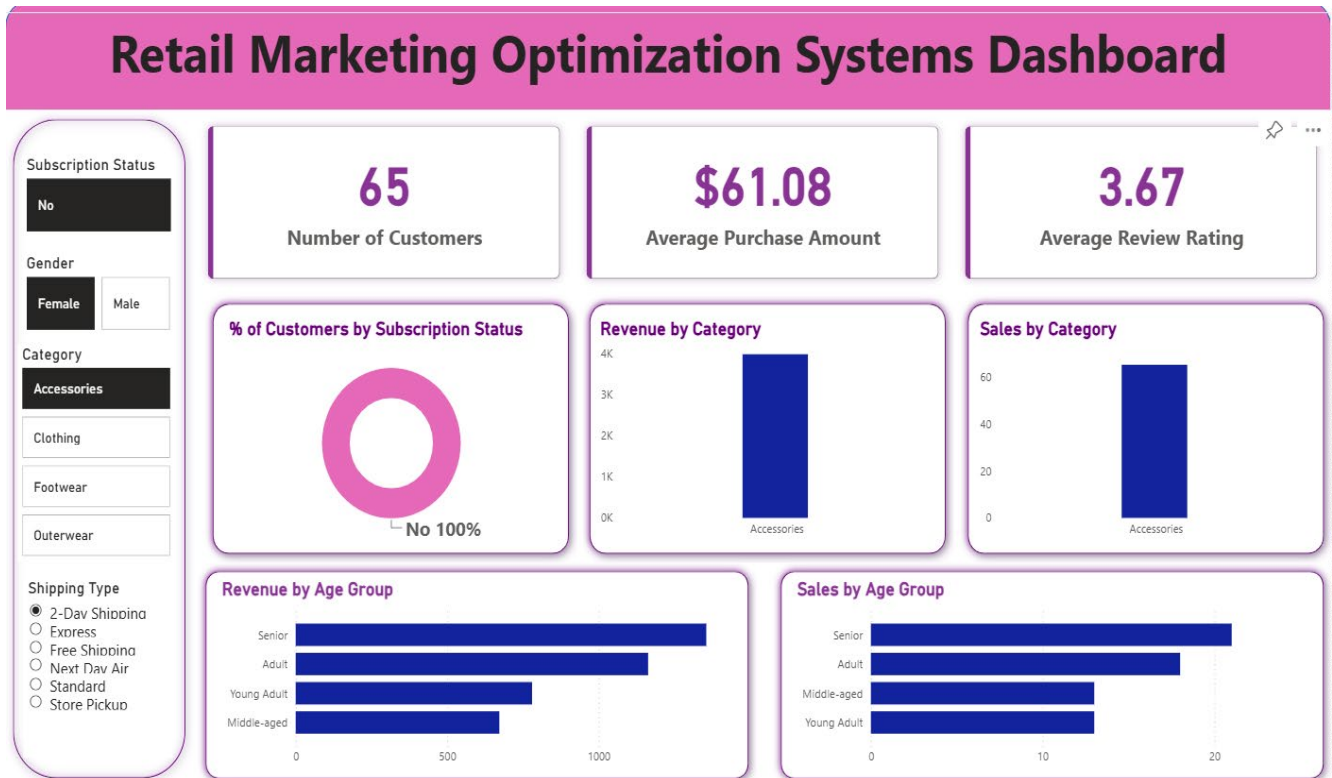
- KPI Cards
- Product Category Sales
- Customer Age Group Distribution
- Payment Method Analysis
- Purchase Trends
- Interactive Filters

These visualizations support quick interpretation of business performance and customer purchasing patterns

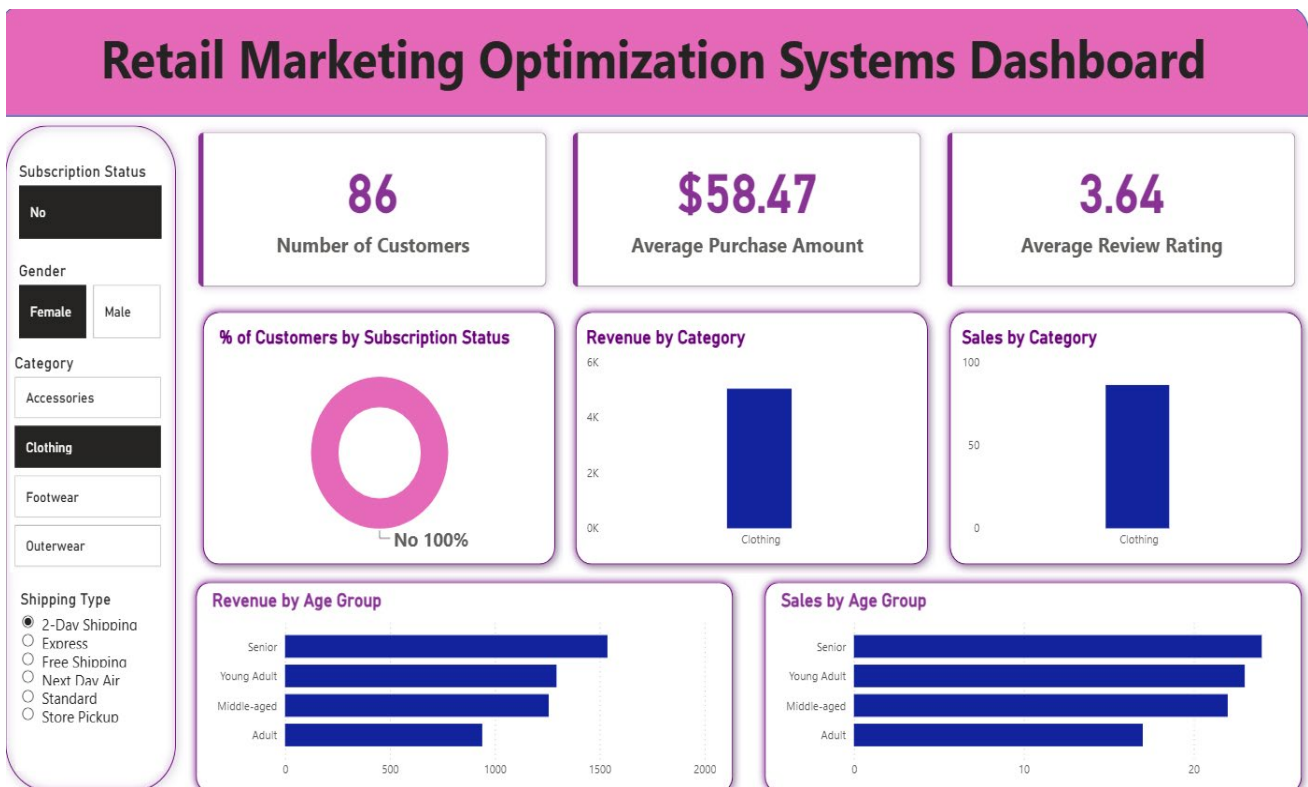
1. Sales as per Subscription Status as Yes and Gender as Male



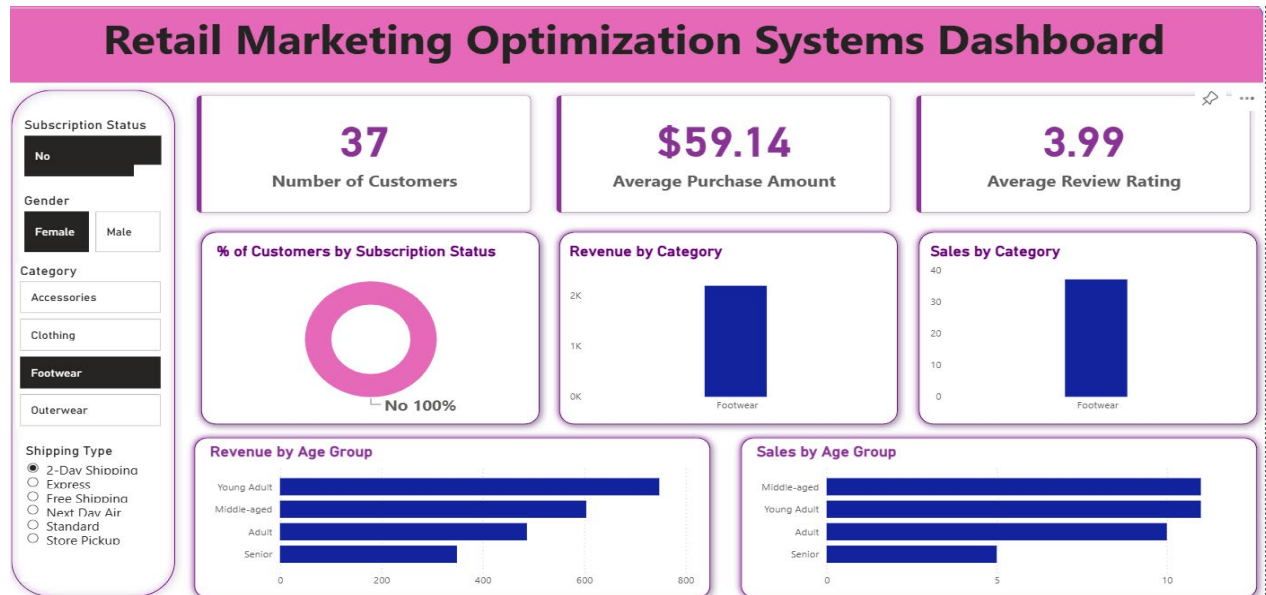
2. Sales as per Subscription Status as No and Gender as Female and Category as Accessories



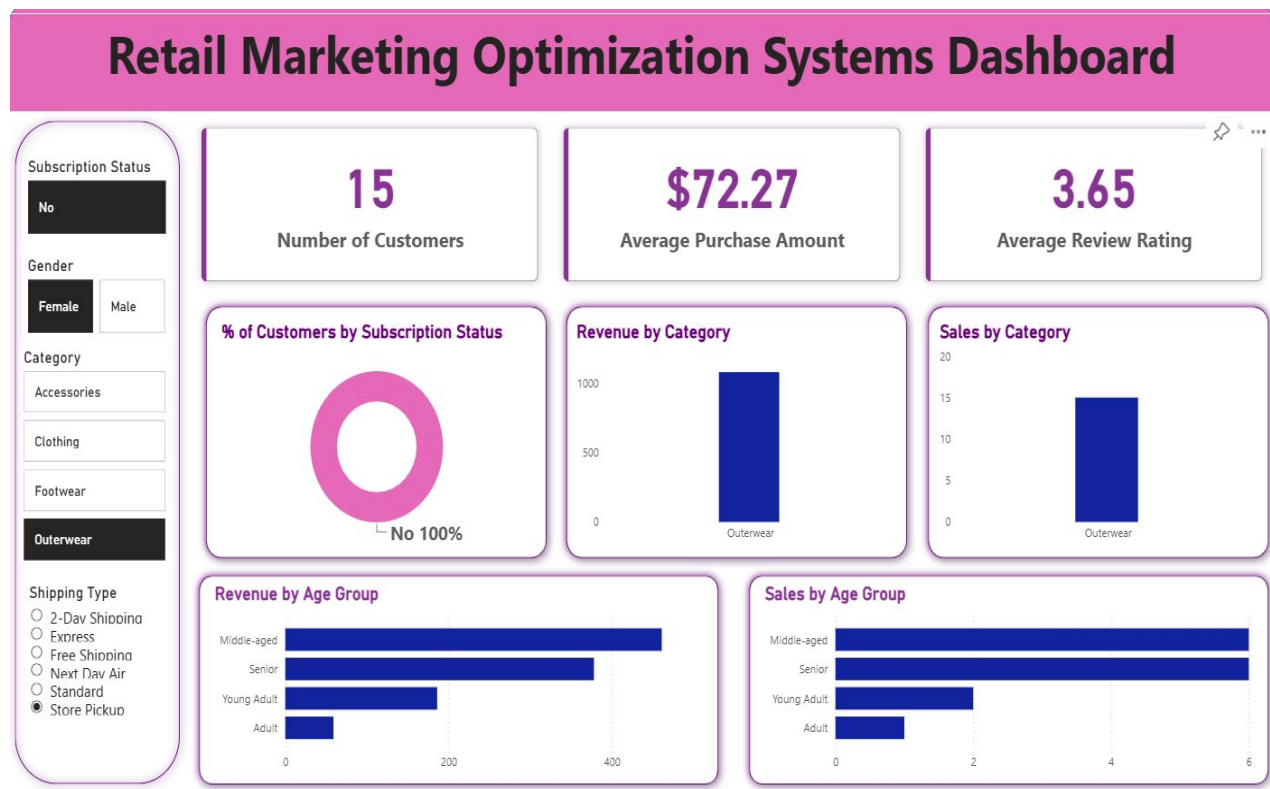
3. Sales as per Subscription Status as No and Gender as Female and Category as Clothing



4. Sales as per Subscription Status as No and Gender as Female and Category as Footwear



5. Sales as per Subscription Status as No and Gender as Female and Category as Outerwear and Shipping as Store Pickup.



6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The **Customer Shopping Behavior Analysis using SQL, Python and Power BI** project demonstrates a complete end-to-end data analytics workflow suitable for modern retail environments. Beginning with raw customer shopping data, the system performs data cleaning, exploratory analysis, database integration, business querying, and interactive visualization.

Python enables efficient preprocessing and statistical analysis, SQL provides reliable storage and retrieval of structured information, and Power BI transforms analytical results into intuitive dashboards. Together, these technologies support informed decision-making by highlighting purchasing trends, customer demographics, product performance, and key business indicators.

The project successfully achieves its primary objectives:

- Data preprocessing and cleaning.
- Exploratory data analysis.
- SQL-based business analytics.
- Interactive dashboard creation.
- Business insight generation.

The integrated approach presented in this project reflects industry-standard practices in data analytics and demonstrates the practical application of multiple technologies to solve real-world business problems.

6.2 Future Scope

The proposed system can be enhanced in several ways to increase its analytical capabilities:

Machine Learning Integration

Implement predictive models for customer segmentation, sales forecasting, and personalized product recommendations.

Real-Time Analytics

Integrate streaming platforms to analyze live transactional data.

Cloud Deployment

Deploy the solution on cloud platforms such as Microsoft Azure, AWS, or Google Cloud to improve scalability and accessibility.

Natural Language Querying

Allow users to ask business questions using natural language, enabling non-technical stakeholders to interact with the system more easily.

Advanced Dashboards

Incorporate additional KPIs, forecasting visuals, and automated alerts for enhanced decision support.

Data Warehouse Integration

Extend the system to support enterprise-scale data warehouses and multi-source data integration.

6.3 Learning Outcomes

This project provides practical experience in:

- Data preprocessing with Python.
- Exploratory Data Analysis (EDA).
- Relational database design and SQL querying.
- Business intelligence dashboard development.
- Data visualization techniques.
- End-to-end analytics workflow implementation.
- Report writing and presentation of analytical findings.

6.4 Final Remarks

The project demonstrates how modern analytics tools can transform raw retail data into meaningful business intelligence. By combining Python, SQL, and Power BI, the proposed system offers a scalable and effective solution for understanding customer shopping behavior and supporting strategic business decisions.

6.5 Appendices

The appendix section will contain:

- Representative Python code snippets.
- Representative SQL queries.
- Power BI dashboard screenshots.
- Dataset schema.
- Additional tables and figures.
- Glossary of key terms.

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